Connected Spaces

Def'n: A space X is connected iff the only subsets of X that are both open and closed in X are X and $\emptyset$.

Lemma: If Y is a subspace of X, a separation of Y is a pair of disjoint nonempty sets A & B whose union is Y, neither of which contains a limit point of the other. The space Y is connected if there is no separation of Y.

Proof: Suppose A & B form a separation of Y. Then A is both open & closed in Y. The closure of A in Y is $\bar{A} \cap Y$ ($\bar{A}$ closure of A in X) Since A closed in Y, $A = \bar{A} \cap Y$. So $\bar{A} \cap B = \emptyset$. Thus B has none of A's limit points. The same logic shows that A has none of B's limit points.

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Suppose that $A, B \subseteq Y$ s.t. $A \cup B = Y$ and $\bar{A} \cap B = \emptyset = A \cap \bar{B}$. Then $\bar{A} \cap Y = A$, so A is closed, and similarly B is closed. But $A = Y \setminus B$ so A is open and so is B. Thus A & B are clopen.

□

Lemma: If C and D form a separation of X and Y is a connected subset of X, then Y is entirely contained in C or D.

Proof: C, D open in X, so $C \cap Y$, $D \cap Y$ are open in Y. Also, because $X = C \cup D$, $Y = (C \cup D) \cap Y = (C \cap Y) \cup (D \cap Y)$. If both these sets were nonempty, then this would be a separation of Y. Thus, at least one is empty, so Y is entirely contained in either C or D.

□

Theorem: The Union of a collection of connected sets that have a point in common is connected.

Proof: Let $\{A_\alpha\}$ be a collection of connected subsets of a space X. Let p be a point of $\cap A_\alpha$. Suppose that $Y = C \cup D$ is a separation of Y. The point p must be in C or D. Suppose $p \in C$. Since A_α is connected, it must be entirely contained in either C or D. But $p \in A_\alpha$, and $p \in C$, so $A_\alpha \subseteq C$. This is true for all A_α, so $\cup A_\alpha \subseteq C$. Thus D is empty. which contradicts C, D being a separation.

□

Theorem: Let $A \subseteq X$ be connected. If $A \subseteq B \subseteq \overline{A}$, then B is connected.

Proof: Let A be connected and $A \subseteq B \subseteq \overline{A}$. Suppose that $B = C \cup D$ is a separation of B. Because A is connected, $A \subseteq C$ or $A \subseteq D$. Suppose $A \subseteq C$. Then D contains a subset of the limit points of A. But Then D contains limit points of C. Thus D must be empty, so C, D is not a separation.

□

Theorem: The image of a connected space under a continuous map is connected.

Proof: Let $f: X \to Y$ be continuous, and X connected. Consider $f|_{f(X)}$, as range restriction methods continuity. call it f^*. Let $f(X) = Z$. Then $f^*: X \to Z$. is surjective and continuous. Suppose $A \cup B$ is a separation of Z. Then

Surjective and continuous. Suppose $A \cup B$ is a separation of Z. Then $f^{-1}(A) \cup f^{-1}(B) = X$. Furthermore, $f^{-1}(A)$ is open and so is $f^{-1}(B)$. Thus $f^{-1}(A) = X \setminus f^{-1}(B)$ is also closed. This means $f^{-1}(A), f^{-1}(B)$ is a separation of X, which is a contradiction. □

Theorem: The Cartesian Product of connected spaces is connected.

Proof: First prove for a product of two connected spaces, $X \times Y$. Let $a \times b \in X \times Y$. $X \times b$ is connected because it is homeomorphic with X, and each $a \times Y$ is connected as they're homeomorphic with Y. As a result, $(X \times b) \cup (X \times Y)$ is connected as it is the union of connected spaces with a point in common, $X \times b$ to be specific. Thus $\bigcup_{x \in X} [(X \times b) \cup (X \times Y)]$ is connected and equals all of $X \times Y$. Thus, $X \times Y$ is connected. Proof for any finite product follows by induction.

We now prove this for an arbitrary product. Let $\{X_\alpha\}_{\alpha \in J}$ be an indexed family of connected spaces, and let $X = \prod_{\alpha \in J} X_\alpha$. Let $b = (b_\alpha)_{\alpha \in J} \in X$. Given a finite set $\{\alpha_1, \ldots, \alpha_n\}$ of indices in J, let $X(\alpha_1, \ldots, \alpha_n)$ be a subspace of X. It is all points $(x_\alpha)_{\alpha \in J}$ such that $x_\alpha = b_\alpha$ if $\alpha \notin \{\alpha_1, \ldots, \alpha_n\}$. We must show that $X(\alpha_1, \ldots \alpha_n)$ is homeomorphic with $X_{\alpha_1} \times \cdots \times X_{\alpha_n}$ which is connected as shown in part 1. Consider the bijection between the spaces mapping $(x_{\alpha_1}, \ldots, x_{\alpha_n}) \to (y_\alpha)_{\alpha \in J}$ by $y_\alpha := x_\alpha$ if $\alpha \in \{\alpha_1, \ldots, \alpha_n\}$, and $y_\alpha = b_\alpha$ otherwise. It follows that this map carries a basis element

by $y_\alpha = x_\alpha$ if $\alpha \in \{\alpha_1, \dots, \alpha_n\}$, and $y_\alpha = b_\alpha$ otherwise. It follows that this map carries a basis element of $X_{\alpha_1} \times \cdots \times X_{\alpha_n}$ to a basis element of $X(\alpha_1, \dots, \alpha_n)$. The space $Y = \bigcup X(\alpha_1, \dots, \alpha_n)$, where the union is over all finite subsets of J, is connected because it is a union of connected spaces that all contain b. Let $(x_\alpha)_{\alpha \in J} \in X$ and $U = \prod U_\alpha$ containing (x_α). We must show that $U \cap Y \neq \emptyset$, for this implies $x_\alpha \in \overline{Y}$, and thus X is connected. Each U_α is open, and $U_\alpha = X_\alpha$ for all but finitely many α, $\{\alpha_1, \dots, \alpha_n\}$. Let $(y_\alpha) \in X$ be defined by

$$y_\alpha = \begin{cases} x_\alpha & \text{if } \alpha \in \{\alpha_1, \dots, \alpha_n\}. \\ b_\alpha & \text{if } \alpha \notin \{\alpha_1, \dots, \alpha_n\}. \end{cases}$$

Then $(y_\alpha) \in Y$ because it is in $X(\alpha_1, \dots, \alpha_n)$. $(y_\alpha) \in U$, because $y_\alpha = x_\alpha$ for all $\alpha \in \{\alpha_1, \dots, \alpha_n\}$, and $b_\alpha \in X_\alpha$ otherwise. Thus, $y_\alpha \in U \cap Y$. Therefore $y_\alpha \in \overline{Y} = X$ is connected. □

Exercises:

2) Show that if $\prod X_\alpha$ is connected and nonempty, then each X_α is connected.

Proof: Consider the map $i_\beta : \prod X_\alpha \to \prod X_\alpha$ defined by

$$i_\beta((x_\alpha)_{\alpha \in J}) = \begin{cases} x_\beta, & \text{if } \alpha = \beta. \\ b_\alpha, & \text{otherwise, where } b_\alpha \text{ fixed.} \end{cases}$$

Let $U \subseteq X_\alpha$ be open. Then $i^{-1}(U)$ is open for the map in each index is either constant or the identity map. Clearly, $i_\beta(\prod X_\alpha)$ is homeomorphic to X_β. Thus, each $i_\beta(\prod X_\alpha)$ is connected by theorem 1.5, because it is the image of a

each $\pi_\alpha(\prod X_\alpha)$ is connected by Theorem 1.5, because it is the image of a connected space under a continuous map. Connectedness of each X_α follows by homeomorphism. □

1) Let τ, τ' be two topologies on X. If $\tau' \supseteq \tau$, what does connectedness in one topology imply about connectedness in the other?

τ' is a finer topology than τ. If τ' is connected, then there is no separation of X in τ'. Suppose τ' is connected but τ is not. Then $\exists A, B \subseteq X$ both open in τ s.t. $X = A \cup B$. But A, B open in τ implies A, B open in τ'. Thus A, B is a separation of X in τ', which is a contradiction. So, if X, τ' is connected, then so is X, τ. If X, τ is connected, X, τ' is not necessarily connected. For example, consider $(0,1) \cup (1,2) \subseteq \mathbb{R}$. Let τ' be the order topology, τ be $\{(0,1) \cup (1,2), \emptyset\}$. X, τ is connected, for there is no open sets A, B necessarily that union to form $(0,1) \cup (1,2)$. But $(0,1) \cup (1,2)$ in the order topology is not connected. □

3) Let $\{A_n\}$ be a sequence of connected subsets of X such that $A_n \cap A_{n+1} \neq \emptyset$ for all n. Show that $\bigcup A_n$ is connected.

Proof: Notice that A_1 and A_2 have a point in common, so $A_1 \cup A_2$ is connected. $A_2 \cap A_3 \neq \emptyset$, so $(A_1 \cup A_2) \cup A_3$ is connected. Define the sequence $\{B_n\} = \bigcup_{i \leq n} A_n$. We must show that each B_n is connected. We proceed by induction. Suppose B_k is connected. Then $B_{k+1} = B_k \cup A_{k+1}$. A_{k+1} is connected and $B_k \supseteq A_k$, so $B_k \cap A_{k+1} \neq \emptyset$. Thus B_{k+1} is connected. We showed above that B_2 is connected, and B_1 is obviously connected, so by induction all B_n [illegible] Notice

Thus B_{n+1} is connected. We showed above that B_2 is connected, and B_1 is obviously connected, so by induction all B_n are connected. Notice that $\bigcup B_n = \bigcup A_n$. Each $B_{n+1} \supseteq B_n$, so they contain points in common, thus $\bigcup B_n$ is connected, and by equality so is $\bigcup A_n$. □

4) Let $\{A_\alpha\}$ be a collection of connected subsets of X; let A be a connected subset of X. Show that if $A \cap A_\alpha \neq \emptyset$ for all α, then $A \cup (\bigcup A_\alpha)$ is connected.

Proof: Notice that $\bigcap A_\alpha$ could be empty. However, each $A_\alpha \cap A$ is not empty. Notice that $A \cup (\bigcup A_\alpha) = \bigcup (A_\alpha \cup A)$. Let $B_\alpha = A_\alpha \cup A$. Then each B_α has the points of A in common and each B_α is connected. Thus $\bigcup B_\alpha$ is connected and so $A \cup (\bigcup A_\alpha)$ is connected. □

5) Show that if X is an infinite set, it is connected in the finite complement topology

$$T_f = \{A \mid X \setminus A \text{ is finite or all of } X\}.$$

Proof: Suppose X is an infinite set with T_f and there is a separation of X. So $A, B \subseteq X$ with $A \cap B = \emptyset$, $A \cup B = X$ and $A, B \in T_f$. Suppose A or B is such that $X \setminus A$ is all of X. Then $A = \emptyset$ and so this is not a separation. Thus, A and B must be s.t. $X \setminus A$, $X \setminus B$ are finite. But $A = X \setminus B$. If B is such that $X \setminus B$ is finite, then $A = X \setminus B$ is finite. If A is finite then $X \setminus A$ cannot be finite, and so A cannot be open. Thus, this is not a valid separation. Therefore there is no separation of X. □

6) A space is totally disconnected if its only connected subsets are one-point sets. Show

6) A space is totally disconnected if its only connected subsets are one-point sets. Show that a finite Hausdorff space is totally disconnected.

Proof: Let X be a finite Hausdorff space, and suppose that A is a connected subset of X. Clearly each one-point set is connected, as there is no separation. We must show that any multi point subset is not connected. Suppose $A = \{x_1, x_2\}$. Then $\{x_1\}, \{x_2\}$ is a separation of A, because $\{x_1\}, \{x_2\}$ are open, disjoint, and $\{x_1\} \cup \{x_2\} = \{x_1, x_2\}$. Suppose A is a finite point set that is disconnected. $A = \{x_1, \dots, x_n\}$. Let $B = \{x_1, \dots, x_i\}$, $C = \{x_{i+1}, \dots, x_n\}$ be a separation of A. Then $A \cup \{y\}$ is disconnected, for $B, C \cup \{y\}$ or $B \cup \{y\}, C$ forms a separation of $A \cup \{y\}$. Thus by induction, any finite set with at least 2 points is disconnected in a Hausdorff space. □

7) Is it true that if X has the discrete topology, then X is totally disconnected? The converse?

Suppose X has the discrete topology. Let $A \subseteq X$. Then let B, C be any 2 sets such that $B \cap C \neq \emptyset$ and $B \cup C = A$, If A is not a singleton. If A is a singleton then of course there is no separation. In the discrete topology, every set is open, so B, C are open and constitute a separation. So yes X is totally disconnected.

Consider $\mathbb{Q}$ under the subset topology of $\mathbb{R}$. Single point sets are not open, but $\mathbb{Q}$ is totally disconnected. So the converse is false.

12) Let $Y \subseteq X$. Let X and Y be connected. Show that if A and B form a separation of $X \setminus Y$, then $Y \cup A$ and $Y \cup B$ are connected.

Proof: $X \setminus Y = A \cup B \Rightarrow (X \setminus Y) \cup Y = (A \cup B) \cup Y$

$\Rightarrow X = (A \cup B) \cup Y$.

Because X is connected, $A \cup B \cup Y$ is connected. $A \cup B \cup Y = (A \cup Y) \cup (B \cup Y)$. A and B are disjoint, so the only points that $(A \cup Y)$ and $(B \cup Y)$ have in common are the points of Y. Furthermore, A and B share no points with Y because $A \cup B = X \setminus Y$. Because $(A \cup B) \cup Y$ is connected, Y is connected, and $(A \cup B) \cap Y = \emptyset$, $A \cup B$ must be connected. For if it were not, then if C, D were a separation of $(A \cup B)$, then $(C \cup Y), D$ would be a separation of $A \cup B \cup Y$.

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